

Hardware Design Document of ML Based Agriculture Robot

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1 SCOPE

This document describes the hardware design and implementation of the system, including the electrical architecture, component selection, power supply design, interfaces, signal connectivity, and PCB design considerations.

2 REFERENCE

Sl.No	Document	Reference
1	System Requirement Document of ML Based Agriculture Robot	RKA_MARS_2V00_SyRD_0V01
2	Block Diagram of ML Based Agriculture Robot	RKA_MARS_2V00_BD_0V01

3 SYSTEM SPECIFICATION

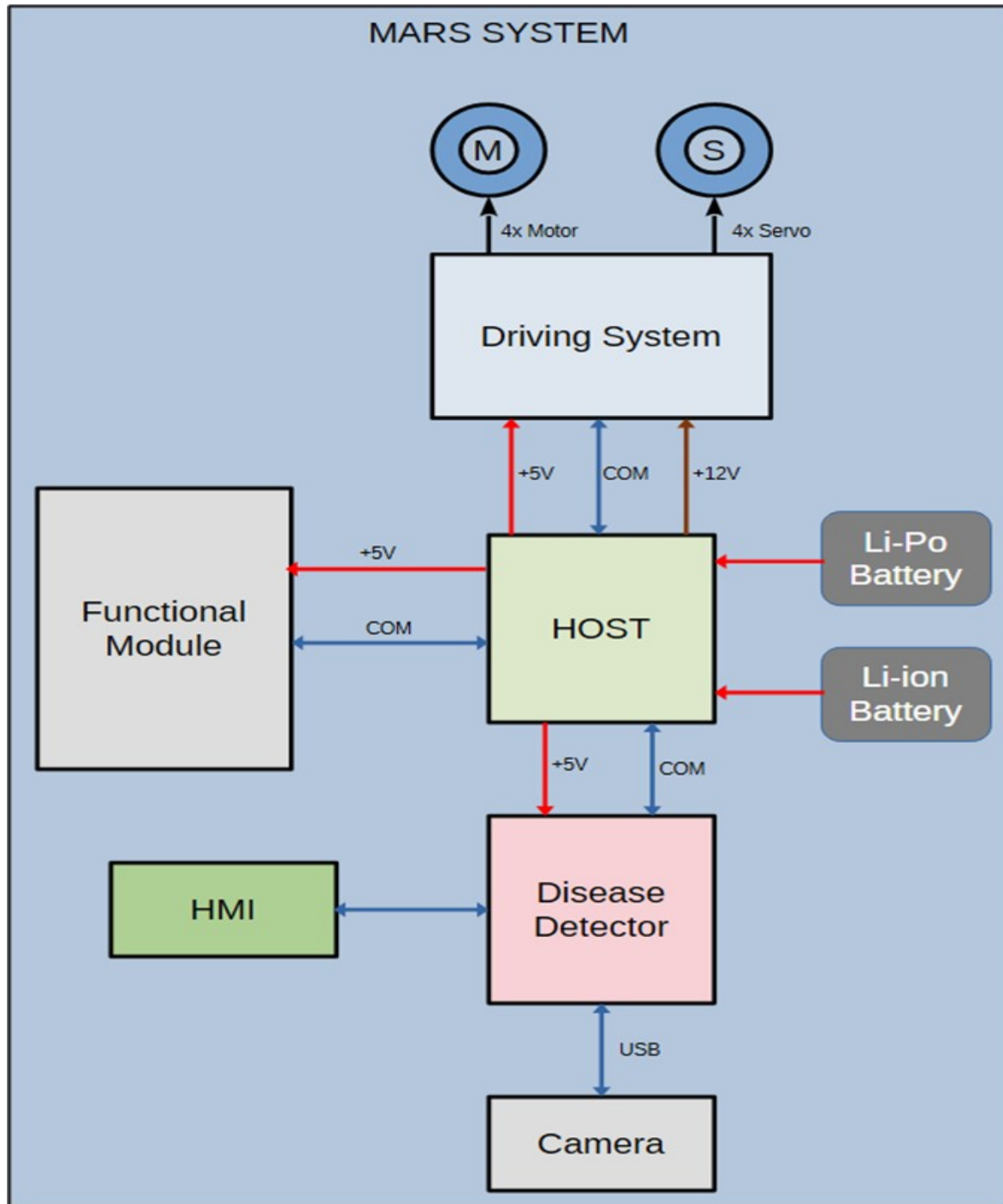
Sl.No	Parameter	Specification
1	Category	Commercial - Proto
2	Stage	Proto
3	Power Supply Source	Internal Battery
4	Operating Voltage	12VDC and 7.2VDC
5	Current Consumption	2A (max) @12VDC and 5A (Max) @ 7.2V
6	Dimension	600 x 450 x 700 mm
7	Weight	Less than 6Kg
8	External Communication	Bluetooth
9	Inter-Module Communication	UART (Modified)
10	Control	Automatic / Semi-Automatic
11	Navigation	GPS / CV
12	ML/DL Model	YOLOv4 or advanced
13	Controlling Element	RP2040 (PICO)
14	Processing Element	Raspberry Pi-5

4 INTRODUCTION

The population of India is highly increasing day by day. But due to lack of manpower and knowledge in the agriculture field the yield and profit are less. Implementation of Robotics and Artificial Intelligence in agriculture will increase the Yield and profit for the farmers. Agriculture work like Fertilizing, Monitoring, and Diseases detection can be done by the single robot using various techniques. Modern Micro-controller and Micro-processor like Arduino and Raspberry Pi can be used for interfacing multiple sensors, actuator, and implementation of Machine Learning. Sensors and Special electrical machines like Stepper motor and Servo motor can be used for accurate result of Weeding and Fertilizing. Machine Learning can be used for Crop monitoring and identification of disease. Implementation of IoT enables the user can be easily monitor crop through Internet in anywhere. It is user (farmer) friendly, and less manpower required.

5 SYSTEM OVERVIEW

6 BLOCK DIAGRAM



7 BLOCK LEVEL DESCRIPTION

1.1 HOST Module

Host Module is the master module which controls all other Modules of the System. The main purpose of the module is two things, Overall System control and Provide the regulated power supply.

Main component of HOST module is Microcontroller, MUX/DEMUX chips, Bluetooth, jumper pin connectors, 5V and 3.3 LDOs

Micro-controller [PICO: RP2040] generates all intermodule control signals and main software sequence are run in it.

MUX/DEMUX chips provide the interface of Micro-Controller Single channel UART can communicate to Multiple Modules.

Bluetooth [HC-05] enables the user can remotely configure, control and get status of system,

Jumper pin connectors, physical layer of interfaces for Intermodule communication.

5V and 3.3 LDO are used provide the power supply to varies components of the system.

Sl.No	Parameter	Specification		
1	Controller Type	ARM (Development Board)		
2	Controller Architecture	ARM-M0+		
3	Input Supply Voltage	12V and 7.8V		
4	Output Supply Voltage	12V	5V	3.3V
5	Output Current	2A	1A	0.5A
6	Wired Interface	1xModified UART, 1xUSB		
7	Wireless Interface	Bluetooth		
8	Dimension	250X150x20 mm		
9	Weight	TDU		
10	Operating Temperature	0 to +55 deg		

1.2 Driving System

Driving System is the Motors driving and its control signal generating module which Drive the Motors of the System based on command received from HOST module. The main purpose of the this module is to drive the DC and Servo motor.

Main component of Driving system is Microcontroller, DC Motor Driver and Servo motor driver.

Micro-controller [PICO: RP2040] generates multiple PWM signals for all motors connected to Driving system.

2 x DC Motor Driver available in Driving System, each Driver can Drive two DC motor independently. So total 4 DC motor can be controllable.

Servo Motor Driver, is customised driver based Open Drain concept using MosFET.

Sl.No	Parameter	Specification
1	Controller Type	ARM (Development Board)
2	Controller Architecture	ARM-M0+
3	Input Supply Voltage	12V and 3.3V
4	Wired Interface	1xUART, 1xUSB
5	Wireless Interface	NA
6	Dimension	125x160x50
7	Weight	TBU
8	Operating Temperature	0 to +55 deg
9	Max no. Of DC Motor port	4 (PWM based speed control)
10	Max no. of Servo Motor port	4 (Customised Driver)

1.3 Disease Detector

Disease Detector is a bought-out processor module, which used to detect disease, Human Machine Interface for User. The main CNN algorithm run on this module. HD Camera, HD Display, Keyboard and mouse will interfaced with this module.

SI.No	Parameter	Specification
1	Processor Type	ARM (Bought-out Board)
2	Controller Architecture	ARM-A33
3	Input Supply Voltage	5V
4	Wired Interface	1x UART, 3x USB, 1x micro HDMI
5	Wireless Interface	NA for system level
6	Dimension	TBU
7	Weight	TBU
8	Operating Temperature	0 to +55 deg
9	Interfacing Peripherals	Display, Camera, Keyboard and mouse
10	Cooling	Forced Air Cooling

1.4 Functional Module

Functional Module is swappable Module, which can swap based required task at that time. For now majorly two Functional Module planned Seeing Module and Fertilizer Sprayer.

1.5 Human Machine Interface

1.6 Camera

1.7 Li-ion Battery

1.8 Li-Po Battery

1.9 Driving Motor

1.10 Steering Servo Motor